EXPERIENCE CERTIFICATE

This is to certify that **Mr. Ankit Raj** was employed as a **Contract Engineer** in the Control Systems Laboratory at **RCI, DRDO**, from **1st May 2025 to 26th April 2026**.

During his tenure, he was associated with the project titled **"FPGA-Based Digital Servo Controller for Electromechanical Actuator"**, where he made valuable contributions to the design verification and validation of the system.

His key roles and responsibilities included:

- Contributed to the verification of an FPGA-based Digital Servo Controller designed for Electromechanical Actuator applications, ensuring reliability and functional accuracy of the system.
- Architected and developed a robust and scalable **UVM (Universal Verification Methodology) testbench** using **SystemVerilog**, incorporating key components such as sequence items, sequences, driver, monitor, agent, and scoreboard to enable modular and reusable verification.
- Designed and executed **constrained-random test scenarios** to thoroughly validate design functionality under diverse operating conditions.
- Developed and implemented **SystemVerilog Assertions (SVA)** for protocol checking and functional validation, enhancing design reliability.
- Performed detailed **functional coverage** and **code coverage (line, toggle, FSM)** analysis, and actively worked towards achieving coverage closure by refining and optimizing test cases.
- Utilized industry-standard EDA tools including **QuestaSim**, **Questa Lint**, and **Xilinx Vivado** for simulation, lint analysis, and FPGA design validation.

Throughout his tenure, Mr. Ankit Raj demonstrated strong technical competence, analytical thinking, and a professional approach towards assigned responsibilities. He exhibited the ability to work effectively both independently and as part of a team.

His performance during his tenure were found to be very good and professional.

We wish him continued success in all his future professional endeavors.